Supplementary material

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1 Appendix A. Reproduction

Every number in this draft is produced by code in scripts/ from artefacts hashed in reports/snapshot.json (230 runs, code, configs, splits, reports). The probes of §4.5 are reports/SATURATION_PROBE.md (pixel statistics), reports/hdr_render_probe.json (soft-shoulder re-render, script scripts/hdr_render_probe.py) and reports/M1_KILL_TEST.md (mask falsification, script scripts/m1_eval.py, per-image predictions in reports/m1_preds_s17,29.npz). The second renderer of §8.4 is reports/PHYSICAL_RENDERER.md, produced by scripts/physical_eval.py from src/maritime_robust/physical.py and gated by scripts/physical_gate.py; its pre-registration is reports/PREREG_PHYSICAL_RENDERER.md. A full re-evaluation of 12 runs reproduced the published R1 statistic bit-exactly (max absolute difference 0). The pre-registration for the severity 2/4 extension is in reports/PREREG_SEVERITY_2_4.md; the analysis and its comparison against the registered predictions is in reports/SEVERITY_CURVE.md. The error-overlap decomposition of §4.6 is reports/COMPOSITION.md §2, from scripts/overlap_main_grid.py; the fifteen-pair composition study and its replication gate are the rest of reports/COMPOSITION.md, from scripts/composition_calibrate,pairs,interaction,predictors,aggregate,gate.py, with the gate pre-registered in reports/PREREG_COMPOSITION_REPLICATION.md. All composition measurements are frozen-model inference; the only training added is six ResNet-50 clean baselines. The coverage study of §5 is pre-registered in reports/PREREG_COVERAGE.md (P1–P13, with the outcome of every prediction recorded there), calibrated by scripts/lattice_calibrate.py, run by scripts/coverage_lattice,singles,singles_hard,singles_mixbase_run.sh, and aggregated into reports/COVERAGE.md by scripts/coverage_aggregate.py; the figure is scripts/coverage_figure.py. The CIFAR-10-C replication of §4.7 is pre-registered in reports/PREREG_CIFAR.md (C1–C5, B1–B3, A1–A3), trained by scripts/cifar_train.py (the AugMix baseline under A1–A3 is scripts/cifar_augmix.sh), evaluated over the 601-condition grid by scripts/cifar_eval.py, validated for port fidelity against the published archive by scripts/cifar_validate.py, and aggregated into reports/CIFAR_GRID.md by scripts/cifar_aggregate.py. The disjoint-augmentation control of §4.7 is

arm otheraug_s17 on the same grid, adjudicated against its frozen predictions O1–O3 by scripts/cifar_otheraug_report.py into reports/OTHERAUG.md.

2 Appendix B. Margin-space superadditivity

The original analysis measured superadditivity in margin space (difference between compound accuracy and the additive expectation of margin losses): haze+glare extra margin loss −0.152 … −0.032, haze+cloud −0.144 … +0.009, across four backbone families. It agrees in sign with the accuracy-space result reported in §4.3 and is not used for inference here.

3 Appendix C. CIFAR-10-C arms: protocol, seeds and intervals

All arms in §4.7 share one protocol: ResNet-18 with the CIFAR stem (3×3, no max-pool), 200 epochs of SGD at learning rate 0.1, momentum 0.9, weight decay 5e-4, nesterov, cosine schedule, batch 128, 4-pixel pad-and-crop and horizontal flip. An arm that trains on degradations replaces the clean view with a corrupted one at probability 0.5 and leaves the geometry pipeline untouched; the clean reference sees clean images only. Every arm is evaluated on the same 601 conditions — clean, 75 single, 525 pair — on the full 10,000-image test set.

arm	seeds	training pool
clean reference	17	none
lattice k=15	17, 29	15 calibrated pairs; the five held-out pairs excluded from every pool
lattice k=40	17	40 calibrated pairs
singles	17, 29	each of the 15 corruption families alone, at the same 0.5 rate
AugMix	17	torchvision AugMix (severity 3) plus the Jensen–Shannon term, $\lambda = 12$

Accuracy at both seeds (percent correct, mean over the conditions of each family):

arm	clean	singles	pairs	held-out s3
clean_s17	95.47	74.77	57.29	61.14
lattice_k15_s17	94.28	91.80	87.88	89.86
lattice_k40_s17	94.16	91.95	88.52	90.16
singles_s17	94.99	92.30	86.77	90.04
lattice_k15_s29	94.31	91.73	87.78	90.08
singles_s29	95.03	92.26	86.58	89.93
augmix_s17	95.36	88.95	78.72	84.84

Intervals. Gain over the clean reference on the 525 pair conditions, with a percentile bootstrap over the 105 corruption pairs (20,000 resamples). The five severities of one pair are not independent draws, so resampling conditions

would overstate precision; resampling pairs keeps them together. The arm contrast is computed per pair first, so its interval is paired.

arm	gain (pp)	95% CI	worst pair	best pair
lattice_k15_s17	+30.60	[+28.47, +32.69]	+4.4	+58.3
lattice_k40_s17	+31.23	[+29.14, +33.32]	+4.7	+58.8
singles_s17	+29.48	[+27.47, +31.50]	+4.8	+57.9
lattice_k15_s29	+30.50	[+28.44, +32.60]	+4.4	+58.4
singles_s29	+29.29	[+27.30, +31.33]	+4.8	+57.8
augmix_s17	+21.43	[+20.19, +22.65]	+4.0	+36.4
augmix_s29	+21.44	[+20.16, +22.73]	+3.9	+38.5

k=15 minus AugMix: +9.17 pp, 95% CI [+8.01, +10.39], higher on 104 of 105 pairs. Seed spreads are 0.10 pp for k=15 and 0.19 pp for singles.

Why raw accuracy and not mCE. mCE normalizes each corruption's error by a fixed AlexNet baseline and aggregates only the fifteen single-corruption families, so it has no slot for pair conditions — the object of this study — and would place our numbers on a scale we did not measure. Arms are therefore compared only against the clean reference trained under the identical recipe, and no claim of parity with published CIFAR-10-C tables is made. The AugMix arm tests the mechanism claim (does generic augmentation substitute for family coverage), not a leaderboard position.

AugMix predictions. Three predictions were frozen before that arm ran. A1: AugMix lands near the singles arm (86.77%) rather than the paired arms (87.88–88.52%). A2: its held-out excess stays positive, unlike the coverage arms (-0.17, -0.42). A3: its clean accuracy does not fall below the clean reference's 95.47%. A1 failed, and in the opposite direction from the prediction: AugMix reaches 78.72%, 8.05 pp below the singles arm. A2 held (+1.97 pp). A3 held to within 0.11 pp (95.36 vs 95.47). The failure is informative rather than awkward: the generic augmentation never targets the families under test, so it does not substitute for covering them, but it also does not approach the arm that shows those families one at a time.

The arm was repeated at a second seed to check that A1's margin is not seed noise. It is not: seed 29 gives 78.73% on compounds against 78.72% at seed 17 (gain over clean +21.44 pp, CI [+20.16, +22.73]), clean 95.85%, singles 89.39%, and the held-out excess stays positive (+2.44 pp at severity 3, against +1.97 pp at seed 17). A generic augmentation that never samples the evaluated families sits ~9 pp below coverage at both seeds, so the +9.17 pp contrast is a property of the arms rather than of one training run — the same reading as the coverage arms' own seed spreads of 0.10 pp and 0.19 pp.

All six frozen predictions on this benchmark.

id	prediction (registered before the run)	result	verdict
B1	held-out excess falls with pool width k	−0.17 (k=15) → −0.42 (k=40)	holds
B2	paired arm beats singles by more than 1 pp	−0.49 (s17), −0.89 (s29)	fails
B3	no arm loses more than 1.5 pp clean accuracy	−0.48 … −1.31	holds
A1	AugMix lands near the singles arm (86.77%)	78.72%, 8.05 pp below it	fails
A2	AugMix's held-out excess stays positive	+1.97 pp	holds
A3	AugMix's clean accuracy not below 95.47%	95.36%	holds (0.11 pp)

Two of six failed, and both failures narrow the paper's claim rather than widen it: B2's failure moved §5 from pair coverage to family coverage, and A1's failure showed that generic augmentation does not substitute for covering the families under test.

4 Appendix D. Intervention table

Candidate	Locus	Mean Δ	95% CI	Verdict
T1 consistency training	training	−0.18	[−1.13, +0.76]	not sustained
I1 conditional normalization	input	+0.94	[−0.09, +1.98]	below criterion
R3 feature denoising	representation	−0.69	[−1.76, +0.35]	no positive evidence
R1 multi-scale fusion	representation	−1.10	[−2.23, +0.05]	interval covers zero
I2 Retinex preprocessing	input	−2.56	[−3.86, −1.22]	negative
T2 group DRO	training	−5.04	[−6.52, −3.54]	negative
T3 curriculum	training	−8.76	[−10.14, −7.40]	negative
T4 ImageNet-22K init	initialization	−0.71	[−2.07, +0.52]	no advantage demonstrated

Three folds × two seeds per candidate, 40 epochs, identical data and corruption instances to §3. Mean Δ is over the 12 compound conditions.

5 Appendix E. Compound-error overlap counts

Condition	compound errors	error under no single at any severity	share
haze+glare, s=2	213	8	3.8%
haze+glare, s=3	241	16	6.6%
haze+glare, s=4	298	57	19.1%
haze+glare, s=5	504	209	41.5%
haze+cloud, s=2	207	2	1.0%
haze+cloud, s=3	217	12	5.5%
haze+cloud, s=4	237	17	7.2%
haze+cloud, s=5	292	67	22.9%

6 Appendix F. Coverage arms

arm	compound sampling	held-out delta (pp)	pool-pair delta (pp)	clean acc (%)
baseline (k=0)	recipe's own compounds	+8.2	+4.1	84.95
c3	3 pairs, all 6 families	−1.8	−0.7	86.25
c6	6 pairs, all 6 families	+2.0	+0.7	86.08
c12	12 pairs, all 6 families	+1.4	−0.2	86.41
s6	6 singles, 20 pp each	+7.2	+4.9	86.73
s6h	6 singles, 40 pp each	+1.9	+1.2	86.08
c12b	12 pairs + base recipe	+4.5	−0.1	85.84
s6b	6 singles + base recipe	+9.7	+5.9	86.33

Three folds × two seeds per arm; every arm applies a compound to half of every batch, so the compound rate is 0.5 everywhere. "Pool pairs" are the pairs a pool contains; the held-out pairs appear in no pool.

References